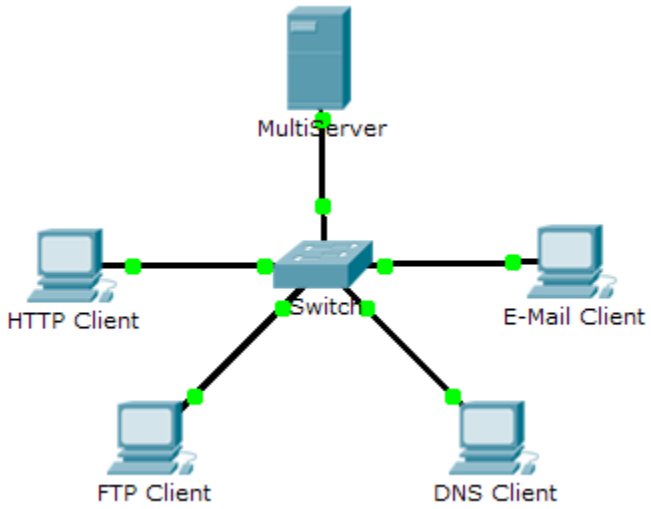
Packet Tracer Simulation - TCP and UDP Communications

Topology



Objectives

Part 1: Generate Network Traffic in Simulation Mode

Part 2: Examine the Functionality of the TCP and UDP Protocols

Background

This simulation activity is intended to provide a foundation for understanding the TCP and UDP in detail. Simulation mode provides the ability to view the functionality of the different protocols.

As data moves through the network, it is broken down into smaller pieces and identified in some fashion so that the pieces can be put back together. Each of these pieces is assigned a specific name (protocol data unit [PDU]) and associated with a specific layer. Packet Tracer Simulation mode enables the user to view each of the protocols and the associated PDU. The steps outlined below lead the user through the process of requesting services using various applications available on a client PC.

This activity provides an opportunity to explore the functionality of the TCP and UDP protocols, multiplexing and the function of port numbers in determining which local application requested the data or is sending the data.

Part 1: Generate Network Traffic in Simulation Mode

Step 1: Generate traffic to populate Address Resolution Protocol (ARP) tables.

Perform the following tasks task to reduce the amount of network traffic viewed in the simulation.

a. Click **MultiServer** and click the **Desktop** tab > **Command Prompt**.

b. Enter the **ping 192.168.1.255** command. This will take a few seconds as every device on the network responds to **MultiServer**.

c. Close the **MultiServer** window.

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Step 2: Generate web (HTTP) traffic.

a. Switch to Simulation mode.

b. Click **HTTP Client** and click the **Desktop** tab > **Web Browser**.

c. In the URL field, enter **192.168.1.254** and click **Go**. Envelopes (PDUs) will appear in the simulation window.

d. Minimize, but do not close, the **HTTP Client** configuration window.

Step 3: Generate FTP traffic.

a. Click **FTP Client** and click the **Desktop** tab > **Command Prompt**.

b. Enter the **ftp 192.168.1.254** command. PDUs will appear in the simulation window.

c. Minimize, but do not close, the **FTP Client** configuration window.

Step 4: Generate DNS traffic.

a. Click **DNS Client** and click the **Desktop** tab > **Command Prompt**.

b. Enter the **nslookup multiserver.pt.ptu** command. A PDU will appear in the simulation window.

c. Minimize, but do not close, the **DNS Client** configuration window.

Step 5: Generate Email traffic.

a. Click **E-Mail Client** and click the **Desktop** tab > **E Mail** tool.

b. Click **Compose** and enter the following information:

   1) **To:** user@multiserver.pt.ptu

   2) **Subject:** Personalize the subject line

   3) **E-Mail Body:** Personalize the Email

c. Click **Send**.

d. Minimize, but do not close, the **E-Mail Client** configuration window.

Step 6: Verify that the traffic is generated and ready for simulation.

Every client computer should have PDUs listed in the Simulation Panel.

Part 2: Examine Functionality of the TCP and UDP Protocols

Step 1: Examine multiplexing as all of the traffic crosses the network.

You will now use the **Capture/Forward button** and the **Back** button in the Simulation Panel.

a. Click **Capture/Forward** once. All of the PDUs are transferred to the switch.

b. Click **Capture/Forward** again. Some of the PDUs disappear. What do you think happened to them?

   The Switch has queued them and put them in its buffer.

c. Click **Capture/Forward** six times. All clients should have received a reply. Note that only one PDU can cross a wire in each direction at any given time. What is this called?

   Multiplexing, mainly Time-Division Multiplexing or serialization. This allows multiple communications (Email, DNS, FTP) to share the same network link by interleaving their packets.

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d. A variety of PDUs appears in the event list in the upper right pane of the simulation window. Why are they so many different colors?

The different colors represent different protocols (e.g., HTTP is purple, DNS is orange, TCP is green)

e. Click **Back** eight times. This should reset the simulation.

Note: Do not click **Reset Simulation** any time during this activity; if you do, you will need to repeat the steps in Part 1.

Step 2: Examine HTTP traffic as the clients communicate with the server.

a. Filter the traffic that is currently displayed to display only **HTTP** and **TCP** PDUs filter the traffic that is currently displayed:

1) Click **Edit Filters** and toggle the **Show All/None** check box.

2) Select **HTTP** and **TCP**. Click anywhere outside of the Edit Filters box to hide it. The Visible Events should now display only **HTTP** and **TCP** PDUs.

b. Click **Capture/Forward**. Hold your mouse above each PDU until you find one that originates from **HTTP Client**. Click the PDU envelope to open it.

c. Click the **Inbound PDU Details** tab and scroll down to the last section. What is the section labeled?

It is labelled as TCP

Are these communications considered to be reliable?

It is reliable as it is sent by TCP protocol

d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is written in the field to the left of the **WINDOW** field?

SRC PORT: 1027, DEST PORT: 80, SEQUENCE NUM: 0, ACK NUM: 0. FLAGS: 0b00000010

e. Close the PDU and click **Capture/Forward** until a PDU returns to the **HTTP Client** with a checkmark.

f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

They are the opposite as now the destination is the client. SOURCE PORT: 80, DEST PORT: 1027. SEQUENCE NUM: 0 as we expect the packet sequence number to be zero based on ack number.

g. There is a second **PDU** of a different color, which **HTTP Client** has prepared to send to **MultiServer**. This is the beginning of the HTTP communication. Click this second PDU envelope and select **Outbound PDU Details**.

h. What information is now listed in the TCP section? How are the port and sequence numbers different from the previous two PDUs?

Now the port numbers are same as the first PDU, as we are again sending to the server. SORCE PORT: 1027 and DEST PORT: 80. But the sequence number is different numbered 1, as ACK was 1 previously

i. Click **Back** until the simulation is reset.

Step 3: Examine FTP traffic as the clients communicate with the server.

a. In the Simulation Panel, change **Edit Filters** to display only **FTP** and **TCP**.

b. Click **Capture/Forward**. Hold your cursor above each PDU until you find one that originates from **FTP Client**. Click that PDU envelope to open it.

c. Click the **Inbound PDU Details** tab and scroll down to the last section. What is the section labeled?

It is labelled as TCP

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Are these communications considered to be reliable?

They are reliable as it is TCP, which is considered reliable

d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is written in the field to the left of the **WINDOW** field?

SRC PORT: 1032, DEST PORT: 21, SEQUENCE NUM: 1, ACK NUM: 53. FLAGS: 0b00010001

e. Close the PDU and click **Capture/Forward** until a PDU returns to the **FTP Client** with a checkmark.

f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

Port numbers changed as the destination previously became source and the destination port is increased by one compared to previous source port. SRC PORT: 21, DEST PORT: 1033. Sequence number is 0

g. Click the **Outbound PDU Details** tab. How are the port and sequence numbers different from the previous two results?

Now the source and destination are the opposite of the inbound details as it is going away, SRC PORT: 1033, DEST PORT: 21. Sequence number is 1 as in the inbound details ACK was 1

h. Close the PDU and click **Capture/Forward** until a second PDU returns to the **FTP Client**. The PDU is a different color.

i. Open the PDU and select **Inbound PDU Details**. Scroll down past the TCP section. What is the message from the server?

Message: Welcome to PT Ftp server

j. Click **Back** until the simulation is reset.

Step 4: Examine DNS traffic as the clients communicate with the server.

a. In the Simulation Panel, change **Edit Filters** to display only **DNS** and **UDP**.

b. Click the PDU envelope to open it.

c. Click the **Inbound PDU Details** tab and scroll down to the last section. What is the section labeled?

It is labelled DNS Query

Are these communications considered to be reliable?

d. Record the **SRC PORT** and **DEST PORT** values. Why is there no sequence and acknowledgement number?

SRC PORT: 1027, DEST PORT: 53. It is UDP, which has no Seq and Ack as it is best effort protocol

e. Close the **PDU** and click **Capture/Forward** until a PDU returns to the **DNS Client** with a checkmark.

f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

They are the opposite of the ones before as the source and destination are interchanged.

g. What is the last section of the **PDU** called?

DNS Answer

h. Click **Back** until the simulation is reset.

Step 5: Examine email traffic as the clients communicate with the server.

a. In the Simulation Panel, change **Edit Filters** to display only **POP3, SMTP** and **TCP**.

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b. Click **Capture/Forward**. Hold your cursor above each PDU until you find one that originates from **E-mail Client**. Click that PDU envelope to open it.

c. Click the **Inbound PDU Details** tab and scroll down to the last section. What transport layer protocol does email traffic use?

TCP

Are these communications considered to be reliable?

It is reliable as TCP is considered reliable.

d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is written in the field to the left of the **WINDOW** field?

SRC PORT: 1027, DEST PORT: 25, SEQUENCE NUM: 0, ACK NUM: 0. FLAGS: 0b00000010

e. Close the **PDU** and click **Capture/Forward** until a PDU returns to the **E-Mail Client** with a checkmark.

f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

The port numbers are interchanged as the destination here is the client. Sequence number is 0 as previously the acknowledgement number was 0, which indicated the next seq num needed. Now ACK: 1

g. Click the **Outbound PDU Details** tab. How are the port and sequence numbers different from the previous two results?

Port numbers are similar to the PDU sent from client at first but opposite to the one received. Sequence number is 1 as previously the acknowledgement number was 1, which indicated the next seq num.

h. There is a second **PDU** of a different color that **HTTP Client** has prepared to send to **MultiServer**. This is the beginning of the email communication. Click this second PDU envelope and select **Outbound PDU Details.**

i. How are the port and sequence numbers different from the previous two **PDU**s?

Port numbers are similar to the PDU sent from client at first but opposite to the one received. Sequence number is 1 as previously the acknowledgement number was 1, which indicated the next seq num.

j. What email protocol is associated with TCP port 25? What protocol is associated with TCP port 110?

k. Click **Back** until the simulation is reset.

Step 6: Examine the use of port numbers from the server.

a. To see TCP active sessions, perform the following steps in quick succession:

1) Switch back to **Realtime** mode.

2) Click **MultiServer** and click the **Desktop** tab > **Command Prompt**.

b. Enter the **netstat** command. What protocols are listed in the left column?

What port numbers are being used by the server?

Protocols: TCP. Port numbers used: 80 (HTTP), 21 (FTP), 25 (SMTP), and 110 (POP3).

c. What states are the sessions in?

ESTABLISHED, TIME_WAIT, or LISTENING.

d. Repeat the **netstat** command several times until you see only one session still ESTABLISHED. For which service is this connection still open? FTP(Port 21)

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Why doesn't this session close like the other three? (Hint: Check the minimized clients)
Because the FTP Client is stateful and we minimized the windows without closing.

Suggested Scoring Rubric

Activity Section	Question Location	Possible Points	Earned Points
Part 2: Examine Functionality of the TCP and UDP Protocols	Step 1	15	
	Step 2	15	
	Step 3	15	
	Step 4	15	
	Step 5	15	
	Step 6	25	
	Total Score	**100**	

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